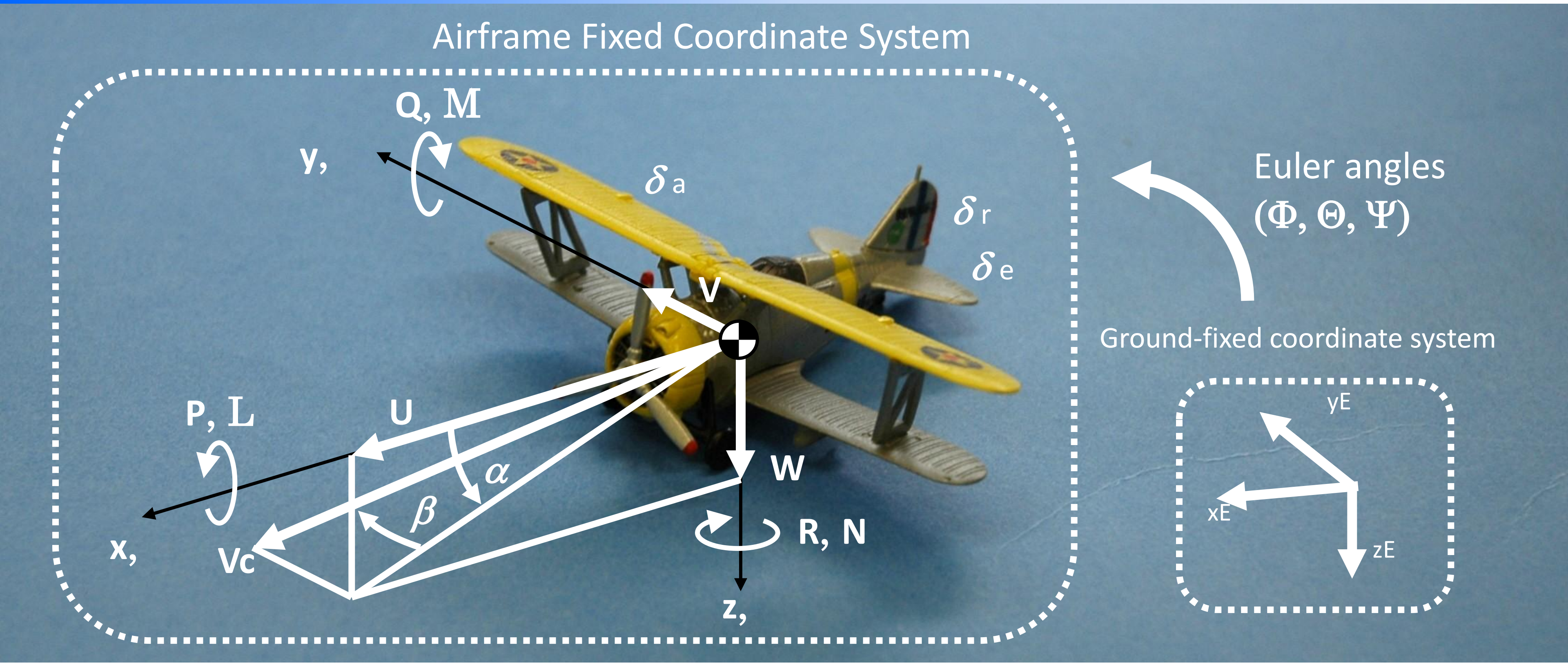
Aeronautical Systems B Lecture: 4

航空宇宙システム特論B　第4回

1. Simulation of aircraft longitudinal dynamics
2. Stability Analysis

1. 航空機の縦運動のシミュレーション
2. 安定解析

Longitudinal dynamics model



x, y, z	: Aircraft-fixed axis coordinate system	α	: Angle of attack.
U, V, W	: x, y, z axis direction velocity	β	: Side slip angle.
P, Q, R	: Angular velocity around the x, y, z axis	Φ, Θ, Ψ	: Euler angle
F_x, F_y, F_z	: External force in x, y, z axis direction	x_E, y_E, z_E	: Ground-fixed coordinate system

Equations of motion for an aircraft expressed in moving coordinate system

$$m(\dot{U} + QW - RV) = -mg\sin\Theta + X_a \quad \text{(A1-47a)}$$

$$m(\dot{V} + RU - PW) = mg\cos\Theta\sin\Phi + Y_a \quad \text{(A1-47b)}$$

$$m(\dot{W} + PV - QU) = mg\cos\Theta\cos\Phi + Z_a \quad \text{(A1-47c)}$$

$$I_{xx}\dot{P} - I_{xz}\dot{R} + (I_{zz} - I_{yy})QR - I_{xz}PQ = L \quad \text{(A1-48a)}$$

$$I_{yy}\dot{Q} + (I_{xx} - I_{zz})RP + I_{xz}(P^2 - R^2) = M \quad \text{(A1-48b)}$$

$$-I_{zx}\dot{P} + I_{zz}\dot{R} + (I_{yy} - I_{xx})PQ + I_{xz}QR = N \quad \text{(A1-48c)}$$

$$P = \dot{\Phi} - \dot{\Psi}\sin\Theta \quad \text{(A1-61a)}$$

$$Q = \dot{\Theta}\cos\Phi + \dot{\Psi}\sin\Phi\cos\Theta \quad \text{(A1-61b)}$$

$$R = -\dot{\Theta}\sin\Phi + \dot{\Psi}\cos\Phi\cos\Theta \quad \text{(A1-61c)}$$

$$V = P = R = 0, \qquad \Phi = 0, \qquad Y_a = L = N = 0$$

$$m(\dot{U} + QW) = -mg\sin\Theta + X_a \quad \text{(B5-1a)}$$

$$m(\dot{W} - QU) = mg\cos\Theta + Z_a \quad \text{(B5-1b)}$$

$$I_{yy}\dot{Q} = M \quad \text{(B5-1c)}$$

$$Q = \dot{\Theta} \quad \text{(B5-1d)}$$

1. Expand the variables and transform it into a first-order differential equation.

(変数を拡張し１階の微分方程式に直す)

$$m(\dot{U} + QW) = -mg\sin\Theta + X_a$$

$$m(\dot{W} + QU) = mg\cos\Theta + Z_a$$

$$I_{yy}\dot{Q} = M$$

$$Q = \dot{\Theta}$$

$$\dot{U} = -QW - g\sin\Theta + X_a/m \qquad \text{(B5-2a)}$$

$$\dot{W} = QU + g\cos\Theta + Z_a/m \qquad \text{(B5-2b)}$$

$$\dot{Q} = M/I_{yy} \qquad \text{(B5-2c)}$$

$$\dot{\Theta} = Q \qquad \text{(B5-2d)}$$

$x = [U \quad W \quad Q \quad \Theta]^T$ State vector (状態ベクトル)

2. Define the differential equation as a function and integrate it using MATLAB's differential equation solver (ode45).

(微分方程式を関数で定義し，matlabの微分方程式ソルバー(ode45)で積分する)

Aerodynamic terms (X_a, Z_a, M) must be modeled as functions of state variables (U, W, Q).

空気力項(X_a, Z_a, M)を状態量(U, W, Q)の関数としてモデル化しなければならない

Longitudinal dynamics model(Without thrusters)

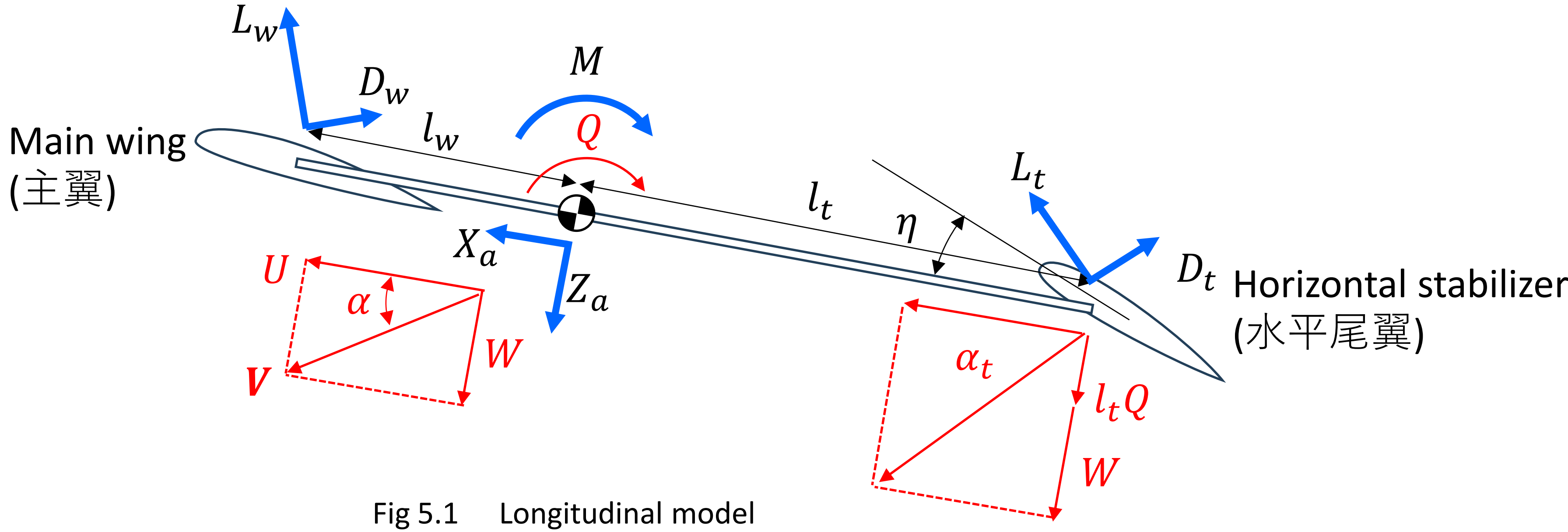


Fig 5.1 Longitudinal model

$$X_a = L_w \sin\alpha + L_t \sin\alpha_t - D_w \cos\alpha - D_t \cos\alpha_t \quad \text{(B5-3)}$$

$$Z_a = -L_w \cos\alpha - L_t \cos\alpha_t - D_w \sin\alpha - D_t \sin\alpha_t \quad \text{(B5-4)}$$

$$M = l_w L_w - l_t L_t \quad \text{(B5-5)}$$

Longitudinal dynamics model

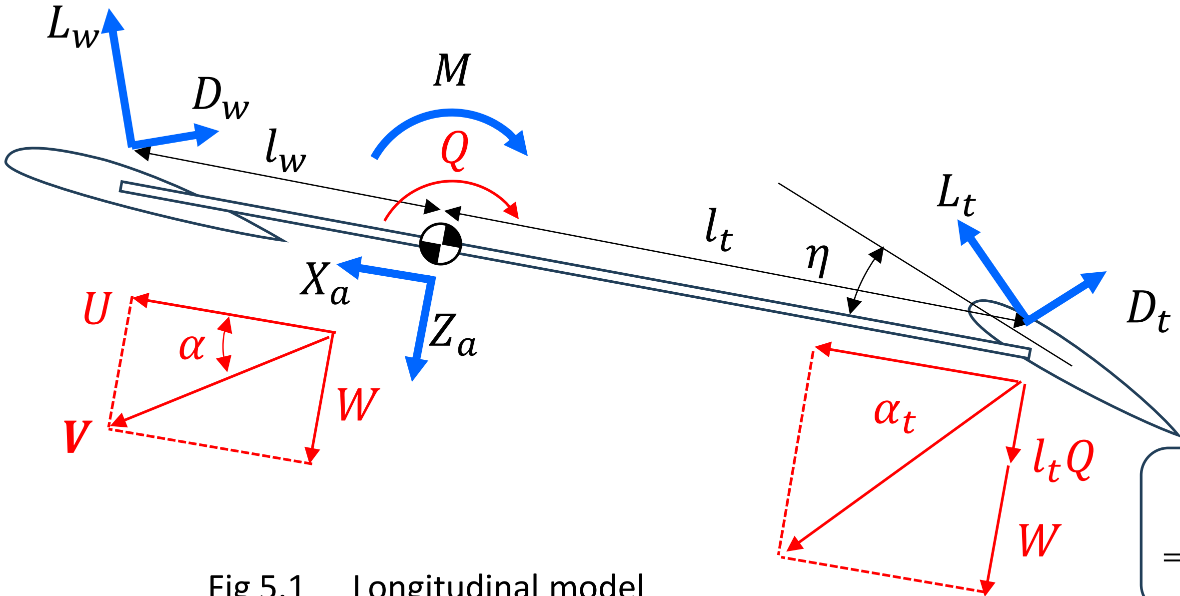


Fig 5.1　Longitudinal model

$$C_D = \frac{C_L{}^2}{\pi A}(1+\delta) + C_{D_{pmin}} + kC_L{}^2$$
$$= C_{D_{pmin}} + \left(k + \frac{1+\delta}{\pi A}\right) C_L{}^2 = C_{D_{pmin}} + \frac{C_L{}^2}{\pi e A} \qquad \text{(B5-30)}$$

Angle of attack of main wing

$$\alpha_w \cong \alpha$$
$$= \tan^{-1}\frac{W}{U} \qquad \text{(B5-6a)}$$

Angle of attack of horizontal stabilizer

$$\alpha_t = \tan^{-1}\frac{W + l_t Q}{U} \qquad \text{(B5-6b)}$$

Lift/Drag of main wing

$$L_w = \frac{1}{2}\rho \boldsymbol{V}^2 S_w C_{L_w} \qquad \text{(B5-7a)}$$

$$D_w = \frac{1}{2}\rho \boldsymbol{V}^2 S_w C_{D_w} \qquad \text{(B5-7b)}$$

Lift/Drag of horizontal stabilizer

$$L_t = \frac{1}{2}\rho \boldsymbol{V}^2 S_t C_{L_t} \qquad \text{(B5-7c)}$$

$$D_t = \frac{1}{2}\rho \boldsymbol{V}^2 S_t C_{D_t} \qquad \text{(B5-7d)}$$

Lift/Drag coefficient of main wing

$$C_{L_w} = C_{L_{w0}} + a_w \alpha_w \qquad \text{(B5-8a)}$$

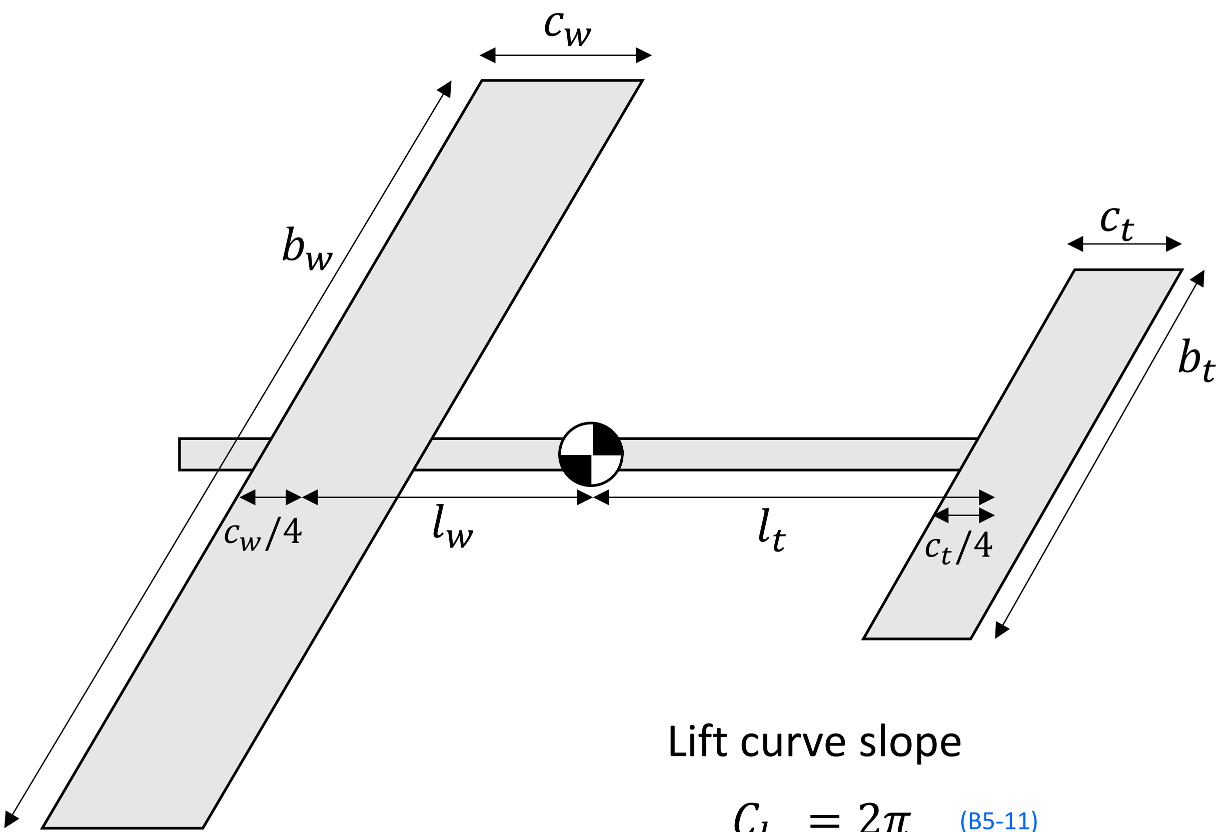
$$C_{D_w} = C_{D_{w0}} + \frac{C_{L_w}{}^2}{\pi e_w A_w} \qquad \text{(B5-8b)}$$

Lift/Drag coefficient of horizontal stabilizer

$$C_{L_t} = C_{L_{t0}} + a_t(\alpha_t + \eta) \qquad \text{(B5-8c)}$$

$$C_{D_t} = C_{D_{t0}} + \frac{C_{L_t}{}^2}{\pi e_t A_t} \qquad \text{(B5-8d)}$$

Example



Lift curve slope

$$C_{l_\alpha} = 2\pi \quad \text{(B5-11)}$$

$$a_w = C_{l_\alpha}\frac{A_w}{2+\sqrt{4+{A_w}^2}} \quad \text{(B5-12a)}$$

$$a_t = C_{l_\alpha}\frac{A_t}{2+\sqrt{4+{A_t}^2}} \quad \text{(B5-12b)}$$

Rectangular flat plate wing

$$C_{L_{w0}} = C_{L_{t0}} = 0 \quad \text{(B5-13)}$$

Drag(Bold approximation)

$$e_w = e_t = 1 \quad \text{(B5-14)}$$

$$C_{D_{w0}} = C_{D_{t0}} = 0.002 \quad \text{(B5-15)}$$

Aspect ratio:

$$A_w = \frac{{b_w}^2}{S_w} \quad \text{(B5-10a)}$$

$$A_t = \frac{{b_t}^2}{S_t} \quad \text{(B5-10b)}$$

$$S_w = b_w c_w \quad \text{(B5-9a)}$$

$$S_t = b_t c_t \quad \text{(B5-9b)}$$

Simulations

Longitudinal_Aircraft.m

```
function dxdt = Longitudinal_Aircraft(t,x)

U = x(1);
W = x(2);
Q =x(3);
Theta = x(4);

g = 9.8; % gravitational acceleration [m/s2]
m = 0.01; % mass [kg] (temporarily value)
Iyy =0.001; % moment of inertia [kgm2] (temporarily value)
lw = -0.01; % Distance between the main wing and the center of gravity [m] (temporarily value)
lt = 0.3; % Distance between the horizontal stabilizer and the center of gravity [m] (temporarily value)
bw = 0.4; % span of main wing [m] (temporarily value)
cw = 0.05; % chord length of main wing [m] (temporarily value)
bt = 0.2; % span of horizontal stabilizer [m] (temporarily value)
ct = 0.05; % chord length of horizontal stabilizer [m] (temporarily value)
eta = 1 / 180 * pi;% angle of horizontal stabilizer [rad] (temporarily value)

rho = 1.29; % air density kg/m3
Clalpha = 2 * pi; % 2D lift curve slope

Sw = bw * cw;
St = bt * ct;

Aw = bw^2/Sw;
At = bt^2/St;

aw = Clalpha * Aw / (2 + sqrt(4 + Aw^2));
at = Clalpha * At / (2 + sqrt(4 + At^2));

CLw0 = 0;
CLt0 = 0;

ew = 1;
et = 1;

CDw0 = 0.002;
CDt0 = 0.002;



alpha = atan(W/U);
alpha_w = alpha;
alpha_t = atan((W+lt*Q)/U);

CLw = CLw0 + aw * alpha_w;
CLt = CLt0 + at * (alpha_t + eta);
CDw = CDw0 + CLw^2 / (pi * ew * Aw);
```


```
cw = 0.05; % chord length of main wing [m] (temporarily value)
bt = 0.2; % span of horizontal stabilizer [m] (temporarily value)
ct = 0.05; % chord length of horizontal stabilizer [m] (temporarily value)
eta = 1 / 180 * pi;% angle of horizontal stabilizer [rad] (temporarily value)

rho = 1.29; % air density kg/m3
Clalpha = 2 * pi; % 2D lift curve slope

Sw = bw * cw;
St = bt * ct;

Aw = bw^2/Sw;
At = bt^2/St;

aw = Clalpha * Aw / (2 + sqrt(4 + Aw^2));
at = Clalpha * At / (2 + sqrt(4 + At^2));

CLw0 = 0;
CLt0 = 0;

ew = 1;
et = 1;

CDw0 = 0.002;
CDt0 = 0.002;



alpha = atan(W/U);
alpha_w = alpha;
alpha_t = atan((W+lt*Q)/U);

CLw = CLw0 + aw * alpha_w;
CLt = CLt0 + at * (alpha_t + eta);
CDw = CDw0 + CLw^2 / (pi * ew * Aw);
```


```
CDt = CDt0 + CLt^2 / (pi * et * At);

Lw = 1/2 * rho *(U^2 + W^2) * Sw * CLw;
Lt = 1/2 * rho *(U^2 + W^2) * St * CLt;
Dw = 1/2 * rho *(U^2 + W^2) * Sw * CDw;
Dt = 1/2 * rho *(U^2 + W^2) * St * CDt;

Xa = Lw*sin(alpha) + Lt * sin(alpha_t) - Dw*cos(alpha) - Dt * cos(alpha_t);
Za = -Lw*cos(alpha) - Lt * cos(alpha_t) - Dw*sin(alpha) - Dt * sin(alpha_t);
M = lw * Lw - lt * Lt;
Udot = -Q*W - g * sin(Theta) + Xa/m;
Wdot = Q*U + g * cos(Theta) + Za /m;
Qdot = M / Iyy;
Thetadot = Q;

dxdt = [Udot
Wdot
Qdot
Thetadot];
```

Simulations

DynamicsSimulation.m

```
clear all
tspan = [0 100];
% initial values
U0 = 1;
W0 = 0;
Q0 =0;
Theta0 = 0;

x0 = [U0
W0
Q0
Theta0];

[t,x] = ode45(@Longitudinal_Aircraft,tspan,x0);
subplot(4,1,1);
plot(t,x(:,1));
xlabel('time[s]');
ylabel('U[m/s]');
subplot(4,1,2);
plot(t,x(:,2));
xlabel('time[s]');
ylabel('W[m/s]');
subplot(4,1,3);
plot(t,x(:,3));
xlabel('time[s]');
ylabel('Q[rad/s]');
subplot(4,1,4);
plot(t,x(:,4)/pi*180);
xlabel('time[s]');
ylabel('pitch angle[deg]');
```

Simulations

Command window of matlab

```
>> DynamicsSimulation
```